

INT-332 Practical-5

set up a Continuous Integration (CI) pipeline using GitHub Actions that automatically builds and pushes a Docker image to Docker Hub whenever code is pushed. Create a workflow file that checks out the repository, logs in to Docker Hub using GitHub secrets, build the images from the Dockerfile, tags it as username/app-ci:latest, and pushes it. Also, briefly explain the role of environment variables and secrets in the workflow.

1. Create this workflow file in your project:

➤ `.github/workflows/ci.yml`

2. Add the following configuration:

```
name: Java Maven CI

on:
  push:
    branches:
      - main
  pull_request:
    branches:
      - main

jobs:
  build:

    runs-on: ubuntu-latest

    steps:
      # 1. Checkout source code
      - name: Checkout Repository
        uses: actions/checkout@v4

      # 2. Setup JDK
      - name: Set up JDK 17
        uses: actions/setup-java@v4
        with:
          java-version: '17'
          distribution: 'temurin'

      # 3. Cache Maven dependencies
      - name: Cache Maven packages
        uses: actions/cache@v4
```

```

with:
  path: ~/.m2
  key: ${ runner.os }}-m2-${ hashFiles('**/pom.xml') }}
  restore-keys: |
    ${ runner.os }}-m2

# 4. Install dependencies, compile and run tests
- name: Build and Test with Maven
  run: mvn clean install

# 5. Archive generated JAR/WAR artifacts
- name: Upload Build Artifacts
  uses: actions/upload-artifact@v4
  with:
    name: java-build-artifacts
    path: |
      target/*.jar
      target/*.war

```

3. Checkout Repository

➤ `uses: actions/checkout@v4`

4. Setup JDK

➤ `uses: actions/setup-java@v4`

5. Cache Maven Dependencies

➤ `uses: actions/cache@v4`

6. Build and Test Application

➤ `run: mvn clean install`

7. Upload Build Artifacts

➤ `uses: actions/upload-artifact@v4`

CI-CD Pipeline · Workflow runs · rajyadav23/docker-workflow/actions/workflows/ci-cd.yml

rajyadav23 / docker-workflow

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Actions New workflow

All workflows

CI-CD Pipeline

Management

- Caches
- Attestations
- Runners
- Usage metrics
- Performance metrics

CI-CD Pipeline ci-cd.yml

4 workflow runs

	Event	Status	Branch	Actor
fix backend build CI-CD Pipeline #4: Commit 062b6f0 pushed by rajyadav23	main	Apr 9, 12:18 PM GMT+5:30 12s	...	
Fix Dockerfile name CI-CD Pipeline #3: Commit 5576907 pushed by rajyadav23	main	Apr 8, 12:49 PM GMT+5:30 18s	...	
Fix CI docker compose issue CI-CD Pipeline #2: Commit 75b25ca pushed by rajyadav23	main	Apr 8, 12:44 PM GMT+5:30 7s	...	
Initial CI/CD Docker project CI-CD Pipeline #1: Commit e1d1111 pushed by rajyadav23	main	Apr 8, 12:34 PM GMT+5:30 20s	...	

93°F Sunny 12:31 PM 5/13/2026

Workflow runs · rajyadav23/java-ci-maven/actions

rajyadav23 / java-ci-maven

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Actions New workflow

All workflows

Java CI with Maven

Management

- Caches
- Attestations
- Runners
- Usage metrics
- Performance metrics

All workflows

Showing runs from all workflows

2 workflow runs

	Workflow	Event	Status	Branch	Actor
Update maven.yml Java CI with Maven #2: Commit 41f9647 pushed by rajyadav23	main	Apr 10, 12:47 PM GMT+5:30 21s	...		
Create maven.yml Java CI with Maven #1: Commit adc26fe pushed by rajyadav23	main	Apr 10, 12:33 PM GMT+5:30 25s	...		

93°F Sunny 12:21 PM 5/13/2026